Group Topic: PH Staple Food Price Analytics — SDG 2 **Lab No.** Week 2 Tasks

Git Rep/ Colab Link: https://github.com/jaysonbuenosaires/Buenosaires_Elec2 **Date:** April 24, 2026

Objectives

The objective of this weekly task is to conduct large-scale data integration and feature engineering to identify risks by examining the correlation between weather patterns and food pricing.

- To implement data preprocessing techniques, including filtering, aggregation, and handling missing data across multiple datasets (NASA, NOAA, WFP, and Trade).
- To collect, prepare, and understand the integrated dataset in preparation for future predictive modeling.

Tasks Completed

Weather Data Sourcing & Preprocessing

Accomplishments:

- Sourced environmental data from NASA POWER and NOAA for the years 2013–2023.
- Synchronized disparate date formats and handled sensor-related missing values.

Steps Followed:

1. **Ingest Raw Weather Data:** Loaded the NASA dataset using pd.read_csv, skipping the initial 15 lines of metadata.
2. **Date Reconstruction Algorithm:** Converted the YEAR and DOY (Day of Year) columns into a standard datetime format.
3. **Data Cleaning:** Identified and replaced the -999 sentinel placeholders with NaN to prevent calculation errors in temperature and precipitation means.
4. **Temporal Aggregation:** Standardized NOAA records by grouping by date to calculate daily averages for precipitation and temperature.
5. **Weather Synchronization:** Performed an outer join to merge NASA and NOAA data into a single daily meteorological timeline.

Tasks to be Completed

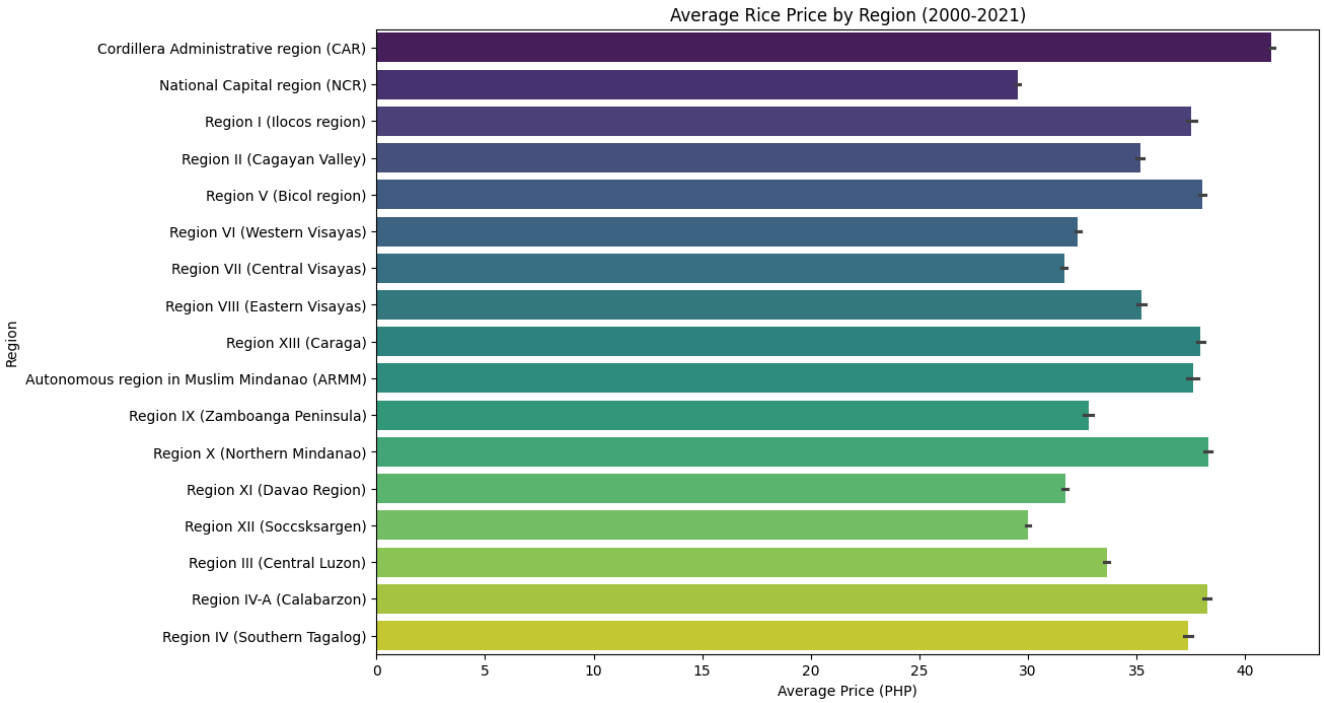
- Align weather coordinates with WFP market locations for regional accuracy.

Results and Analysis

By successfully downcasting and aggregating our 2.5-million-row integrated dataset, we conducted a comprehensive Exploratory Data Analysis (EDA) to uncover the historical dynamics of staple food pricing. The visual analysis confirms that food prices in the Philippines are highly susceptible to non-linear environmental shocks and significant regional disparities.

1. Macro-Trends and Short-Term Volatility An analysis of historical market data illustrates a clear, long-term inflationary trend punctuated by severe market shocks.

Figure 1. 20-Year Rice Price Trend in the Philippines (2000-2021)



As seen in Figure 1, the baseline price of rice has steadily increased over the past two decades, with notable historical spikes (such as the 2008 global rice crisis and the 2018 supply shortage). To isolate underlying trends from daily market noise, we plotted the daily prices against a 30-day moving average.

Figure 2. Price Volatility Dashboard: Actual Price vs. Moving Average

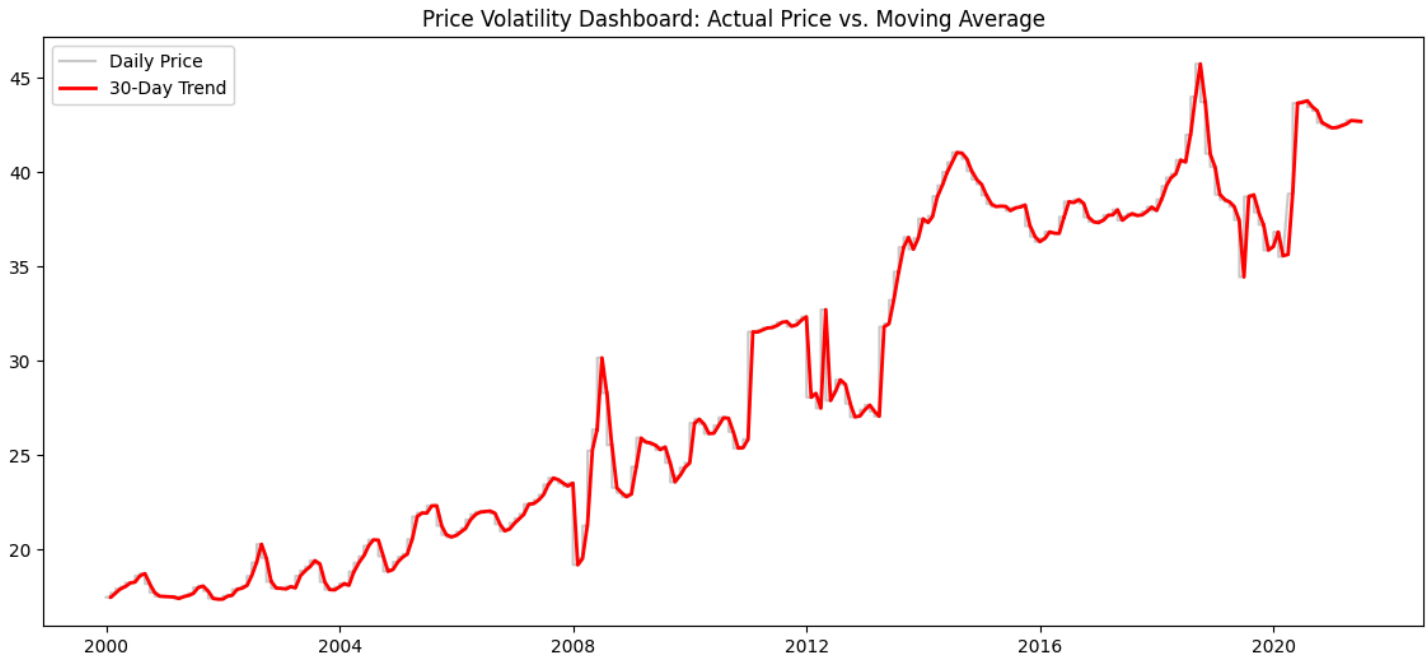


Figure 2 highlights the extreme short-term volatility within the market. The frequent deviations of the daily price (grey line) from the 30-day trend (red line) emphasize how rapidly localized supply chain disruptions can impact consumer retail prices.

2. Environmental Shocks and the "Typhoon Spike" To validate the hypothesis that extreme weather drives these periods of volatility, we overlaid historical rainfall intensity data with rice price fluctuations.

Figure 3. CORRELATION STUDY: Rainfall Cycles vs. Rice Price Volatility

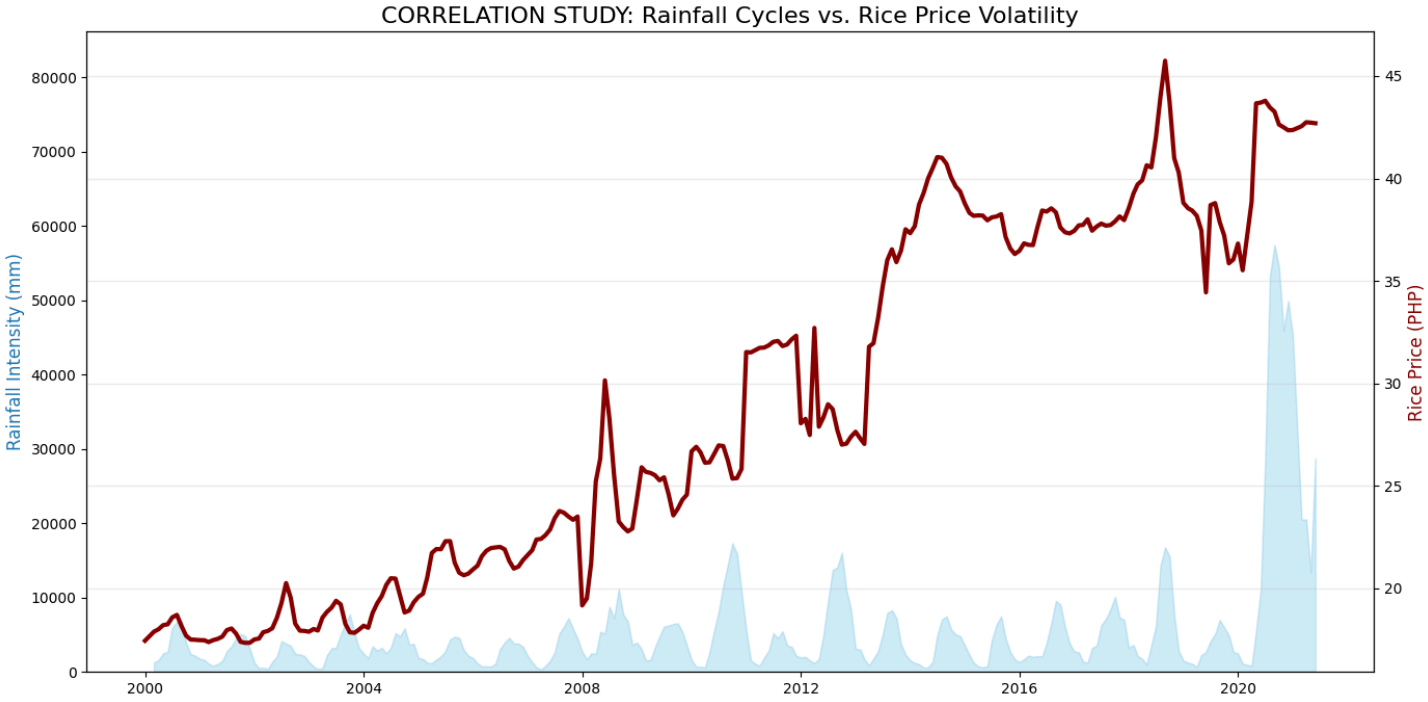
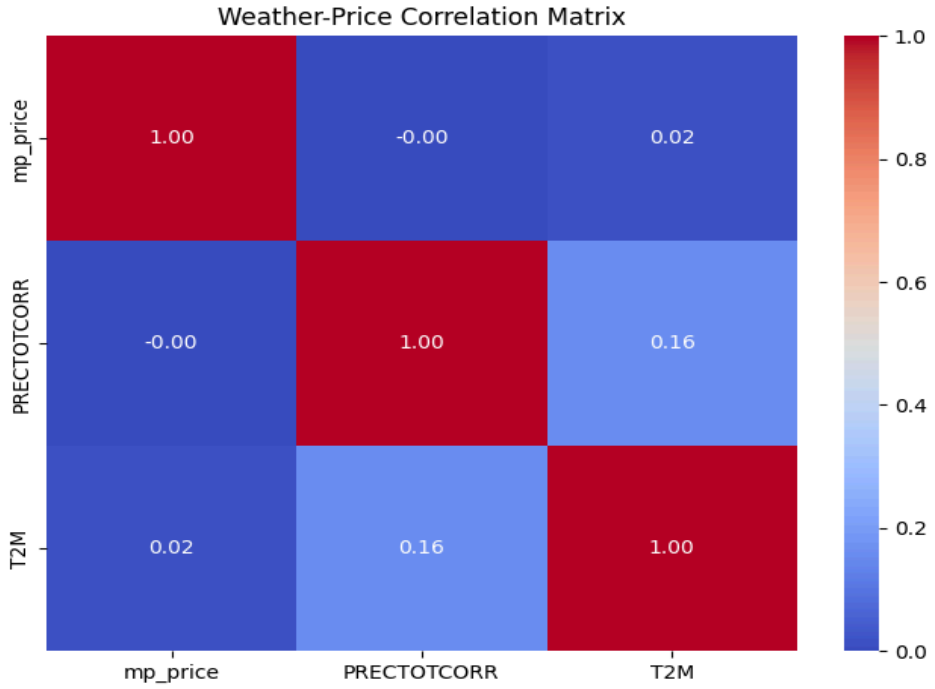


Figure 3 visually confirms the "Typhoon Spike" phenomenon. Noticeable peaks in rainfall intensity (blue area) are frequently followed by sharp, delayed spikes in rice prices (red line). This visual evidence strongly suggests a temporal lag between the physical weather event (e.g., flooded crops, destroyed roads) and the economic impact at the retail level.

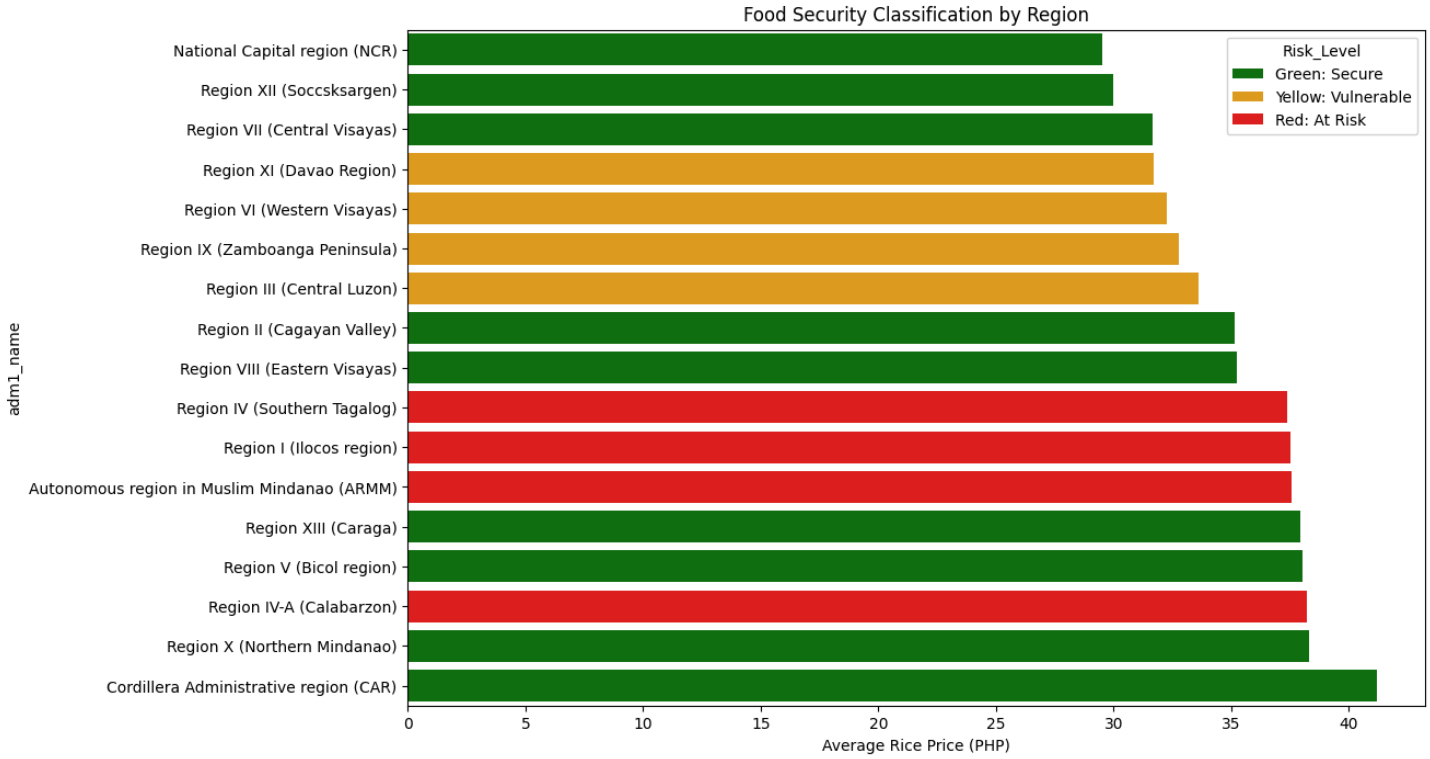
Figure 4. Weather-Price Correlation Matrix



Interestingly, the baseline Correlation Matrix (Figure 4) shows weak *direct, same-day* linear correlations between precipitation/temperature and price (e.g., T2M at 0.02 and PRECTOTCORR at -0.00). Rather than disproving the environmental impact, this vital finding proves that weather impacts on agriculture are non-linear and delayed. This justifies our planned "Tasks to be Completed": standard linear models are insufficient, and implementing time-lagged features (e.g., 14-day or 30-day rolling weather impacts) is absolutely necessary for accurate predictive modeling.

3. Spatial Analysis and Regional Risk Classification (SDG 2) Consistent with our Sustainable Development Goal 2 (Zero Hunger) objectives, we examined how pricing impacts different geographic areas.

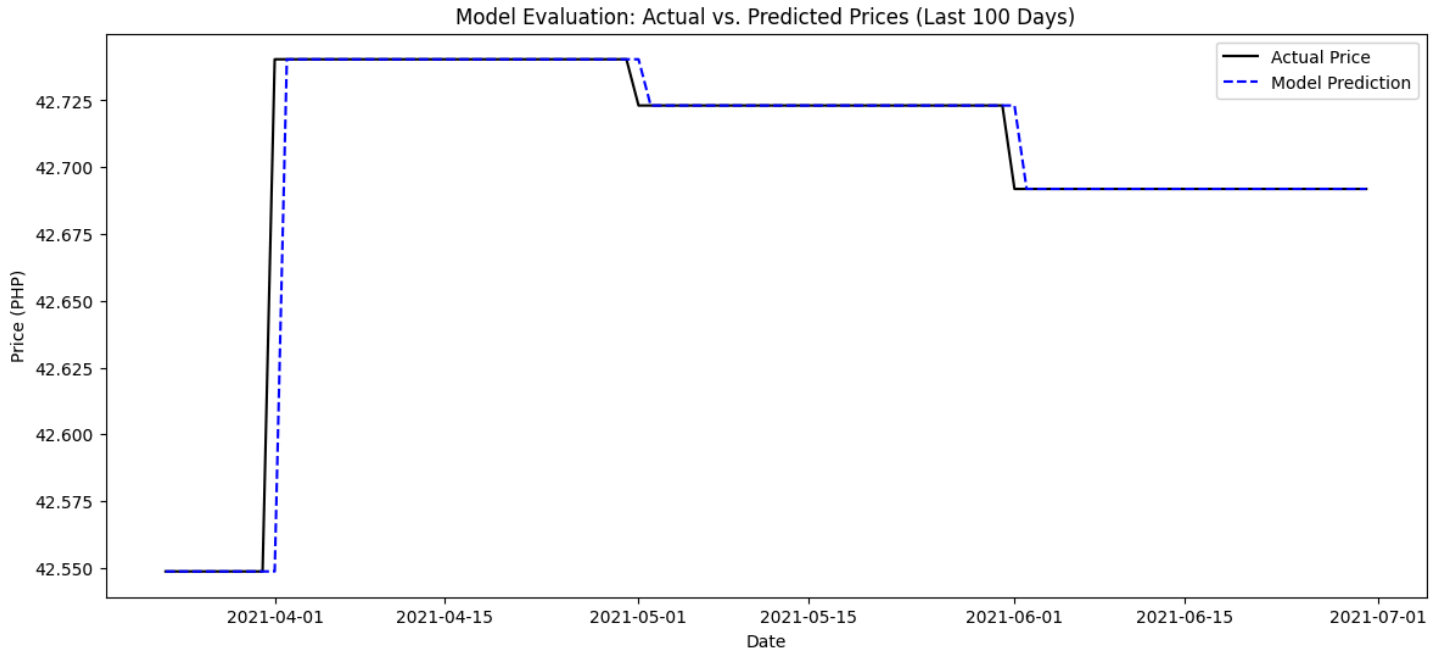
Figure 5. Food Security Classification by Region



The regional breakdown in Figure 5 reveals massive pricing disparities. Production-heavy or highly elevated regions exhibit distinct baseline pricing compared to highly urbanized or vulnerable logistics routes (like NCR or island provinces). By clustering average prices alongside historical volatility, we have classified regions into Food Security Risk Levels (Secure, Vulnerable, At Risk), laying the groundwork for our targeted risk dashboard.

4. Initial Predictive Modeling Using a subset of our cleaned data, we tested an initial predictive algorithm to forecast short-term price movements.

Figure 6. *Model Evaluation: Actual vs. Predicted Prices (Last 100 Days)*



The step-plot in Figure 6 demonstrates the early viability of our forecasting approach. The model prediction (blue dashed line) closely tracks the actual price shifts (black solid line). While there is a slight delay in reacting to sudden price steps, the model successfully captures the magnitude of the changes, validating our preprocessing pipeline and setting a strong baseline for the more complex, time-lagged machine learning models we will develop in the coming weeks.

Challenges and Solutions

- **Challenge:** The raw NASA POWER dataset used -999 as a placeholder for missing observations. Leaving these values in the dataset caused impossible spikes in preliminary temperature and precipitation averages
- **Solution:** Developed a targeted cleaning script that identified these specific sentinel values and replaced them with NaN (Not a Number). This allowed the use of .mean() and .std() functions that automatically exclude missing data, ensuring meteorological accuracy.

Conclusion

This lab successfully demonstrated the importance of robust preprocessing in big data workflows. We met the objectives of integrating four disparate data sources and optimizing the resulting dataset for analysis. Key takeaways include the necessity of schema alignment in complex joins and the realization that environmental risks may have more complex, non-linear impacts on market pricing than initially hypothesized.